Seamoon Pandey

B.E. Computer Engineering | AI & Backend Engineer

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Tansen-14, Palpa, Nepal | Date of birth: 27 January 2003 | Nationality: Nepali

PROFILE

AI and backend engineer working on machine learning pipelines, scalable microservices, and production systems. Currently building Web3 products at Lamida Inc. and applying for an M.Sc. in Computer Science.

EDUCATION

Pashchimanchal (Western Region) Campus, Institute of Engineering, Tribhuvan University — Pokhara, Nepal

Bachelor of Engineering in Computer Engineering — 2022 – 2026

Aggregate: **67.98%**, **First Division** | 1096.1 / 1612.5 marks | T.U. Regd. 3-2-45-173-2021

Tribhuvan University scale: Distinction $\geq$ 80%, First Division $\geq$ 65%, Second Division $\geq$ 50%. Examinations are set and moderated centrally by the university and marks are not scaled.

Thesis: RedSentinel, an AI-assisted cross-site scripting (XSS) vulnerability scanner. Lead architect, team of 3.

LANGUAGE PROFICIENCY

- **Nepali:** native speaker. **Medium of instruction:** English, throughout the Bachelor's degree.

THESIS PROJECT

RedSentinel: AI-Assisted XSS Vulnerability Scanner — Final-year Thesis, Team of 3 | Code | Report | Demo

Lead architect; owned the NestJS orchestration core and the end-to-end ML pipeline. Teammates built the fuzzer and obfuscation engines.

- Fine-tuned a **DistilBERT** multi-task classifier to label an injection point's context and severity, replacing brittle signature matching, and reached near-perfect accuracy on an **8,869-sample** held-out split of a deterministic, well-structured label space.
- Built the full ML pipeline on a **59,122-entry** curated XSS dataset: data curation, custom tokenizer training, multi-task training with checkpointing, evaluation, and an inference API with ONNX/TorchScript export and confidence calibration.
- Architected a polyglot microservice system: a **TypeScript**/**NestJS** core orchestrating three **Python**/**FastAPI** AI services over a **Redis**/**BullMQ** job queue, with **PostgreSQL** persistence and a real-time **Next.js** dashboard (Socket.IO).
- Co-developed an **XGBoost** payload ranker (35 engineered features) to prioritise payloads by predicted effectiveness, with a heuristic fallback when the trained artifact is unavailable.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, Ruby, C/C++, Dart
- **AI / ML:** PyTorch, Transformers (DistilBERT), XGBoost, YOLOv8, ONNX/TorchScript, feature engineering, model evaluation
- **Backend:** Node.js, NestJS, FastAPI, Express.js, Flask, Ruby on Rails, REST, WebSocket/Socket.IO
- **Data & Infra:** PostgreSQL, Redis, BullMQ, Docker, Docker Compose, microservices, async job queues
- **Frontend & Security:** React, Next.js, Flutter; XSS detection, fuzzing, WAF analysis, headless-browser verification (Playwright)
- **Practices:** system architecture, MVC, auth (JWT/API keys), e2e/unit testing, Git, Linux

SELECTED PROJECTS

- **AI Artifact Recognition Platform** (PEC Hackathon: **YOLOv8**, Flask, React). Flask backend and React dashboard for real-time object recognition, with asynchronous inference and structured monitoring endpoints.
- **Community Connect** (OSM Hackathon: Express.js, Flutter). Full-stack event-coordination platform with REST APIs for real-time GPS-based attendance validation and event tracking.
- **Garva Diary** (BMC InnovateX: Ruby on Rails). Backend APIs on a Rails MVC architecture with structured validation workflows and mobile integration.

EXPERIENCE

Full-Stack Engineer, Lamida Inc. — Full-time | 23 Feb 2026 – Present

- Build full-stack Node.js applications for scalable Web3 products, including a DAO for the tUSDT token.
- **Team Lead, Bitfolio** from September 2026.

Freelance Full-Stack Developer — Contract, Remote

- Delivered production platforms on an **Express.js + React** stack (authentication, REST APIs, third-party integrations) while carrying a full course load: **Jyotisam**, an astrology consultation and booking platform (jyotisam.com), and **Mahadaan**, a donation and charity platform with secure data handling (mahadaan.com).

Mentor, Call for Enthusiasts Programme — Innovative Computer Engineering Students' Society (i-CES)

- Mentored junior students on scalable application design, clean backend architecture, and development practice.